

SL-GPS Validation Report: Methyl Formate (CH₃OCHO) Mechanism Reduction via Supervised Learning — ThInK 1.0

SL-GPS Automated Validation Framework

September 21, 2026

Abstract

This report presents a validation of the Supervised Learning – Global Pathway Selection (SL-GPS) method for adaptive chemical mechanism reduction applied to methyl formate (CH₃OCHO) combustion using the ThInK 1.0 mechanism (160 species, 1089 reactions). The complete pipeline is documented: training data generation via GPS, neural network training, and validation against the detailed mechanism across 8 test cases spanning temperatures of 800–1500 K, pressures of 1–20 atm, and equivalence ratios of 0.7–1.3. All training parameters, species classifications, and results are reported for full reproducibility.

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1 Introduction

The SL-GPS method trains neural networks on Global Pathway Selection (GPS) data to dynamically predict which chemical species are important during combustion simulations. At each timestep, the ANN evaluates the current thermochemical state and activates only the predicted species, producing an adaptive reduced mechanism that varies in size throughout the simulation.

This report validates SL-GPS on the ThInK 1.0 mechanism [1] for methyl formate (CH_3OCHO , also known as methyl methanoate), a simple oxygenated fuel of interest as a biodiesel surrogate and a key intermediate in dimethyl ether (DME) and dimethoxymethane (DMM) oxidation.

1.1 Mechanism Overview

Table 1: ThInK 1.0 mechanism summary.

Property	Value
Name	ThInK 1.0
Species	160
Reactions	1089
Target fuels	$\text{C}_1\text{--C}_3$ oxygenated fuels
Format	Cantera YAML

2 Methodology

2.1 SL-GPS Pipeline

The pipeline consists of three stages:

1. **Data generation:** Run N autoignition simulations with randomized initial conditions (T , P , ϕ). At each timestep interval, apply GPS ($\alpha = 0.001$) to label which species are important.
2. **ANN training:** Train an ensemble of neural networks on the GPS-labeled data. Select the best model by validation loss.
3. **Validation:** Run new autoignition cases with the detailed mechanism and simultaneously with SL-GPS (ANN-guided reduction). Compare temperature, HRR, species profiles, and ignition delay.

2.2 Species Classification

Before training, species are partitioned into three groups based on occurrence frequency in the GPS training data:

- **Always** ($> 99\%$ occurrence): unconditionally included in all reduced mechanisms
- **Variable** ($1\text{--}99\%$): predicted by the ANN at each timestep
- **Never** ($< 1\%$ occurrence): permanently excluded

Only the *variable* species are outputs of the neural network.

2.3 Neural Network Architecture

- **Input:** temperature, pressure, and 14 selected species mole fractions (normalized via MinMaxScaler)
- **Hidden layers:** 2 fully-connected layers $\times$ 32 neurons with ReLU activation
- **Output:** sigmoid activation (one output per variable species)
- **Loss:** binary cross-entropy; **Optimizer:** Adam
- **Training:** ensemble of 8 models in parallel, best selected by validation loss
- **Early stopping:** patience = 30 epochs, max = 200 epochs

3 CH₃OCHO — ThInK 1.0

3.1 Training Configuration

Table 2: CH₃OCHO training configuration summary.

Parameter	Value
<i>Mechanism</i>	
Mechanism	ThInK 1.0
Species / Reactions	160 / 1089
<i>Data Generation</i>	
Training cases	15 (parallel execution)
Temperature range	800–1600 K (uniform)
Pressure range	1–20 atm (log-uniform)
Equivalence ratio ϕ	0.7–1.4 (uniform)
GPS α	0.001
Total training samples	707
<i>Species Classification</i>	
Always species	14 (unconditionally included)
Variable species (ANN outputs)	63
Never species	83 (permanently excluded)
<i>ANN Training</i>	
Input features	T, P + 14 species mole fractions
Architecture	2 hidden layers $\times$ 32 neurons
Ensemble size	8 parallel trainings
Max epochs / Early stopping	200 / patience 30

Input species (ANN features): CH₃OCHO, O₂, H₂O, CO₂, CO, OH, H, H₂, CH₄, CH₂O, CH₃OH, HO₂, CH₃O, HCO

3.2 Validation Cases

Table 3: CH₃OCHO validation conditions (4 baseline + 4 extended).

Case	T_0 (K)	P (atm)	ϕ	t_{end} (s)
T1000_P1	1000	1	1.0	0.1
T1200_P1	1200	1	1.0	0.01
T1200_P10	1200	10	1.0	0.005
T1500_P1	1500	1	1.0	0.002
<i>Extended cases (difficult conditions)</i>				
T800_P20	800	20	1.0	0.5
T900_P10	900	10	1.0	0.5
T1100_P1_rich	1100	1	1.3	0.05
T1400_P5_lean	1400	5	0.7	0.01

3.3 Results — Baseline Cases

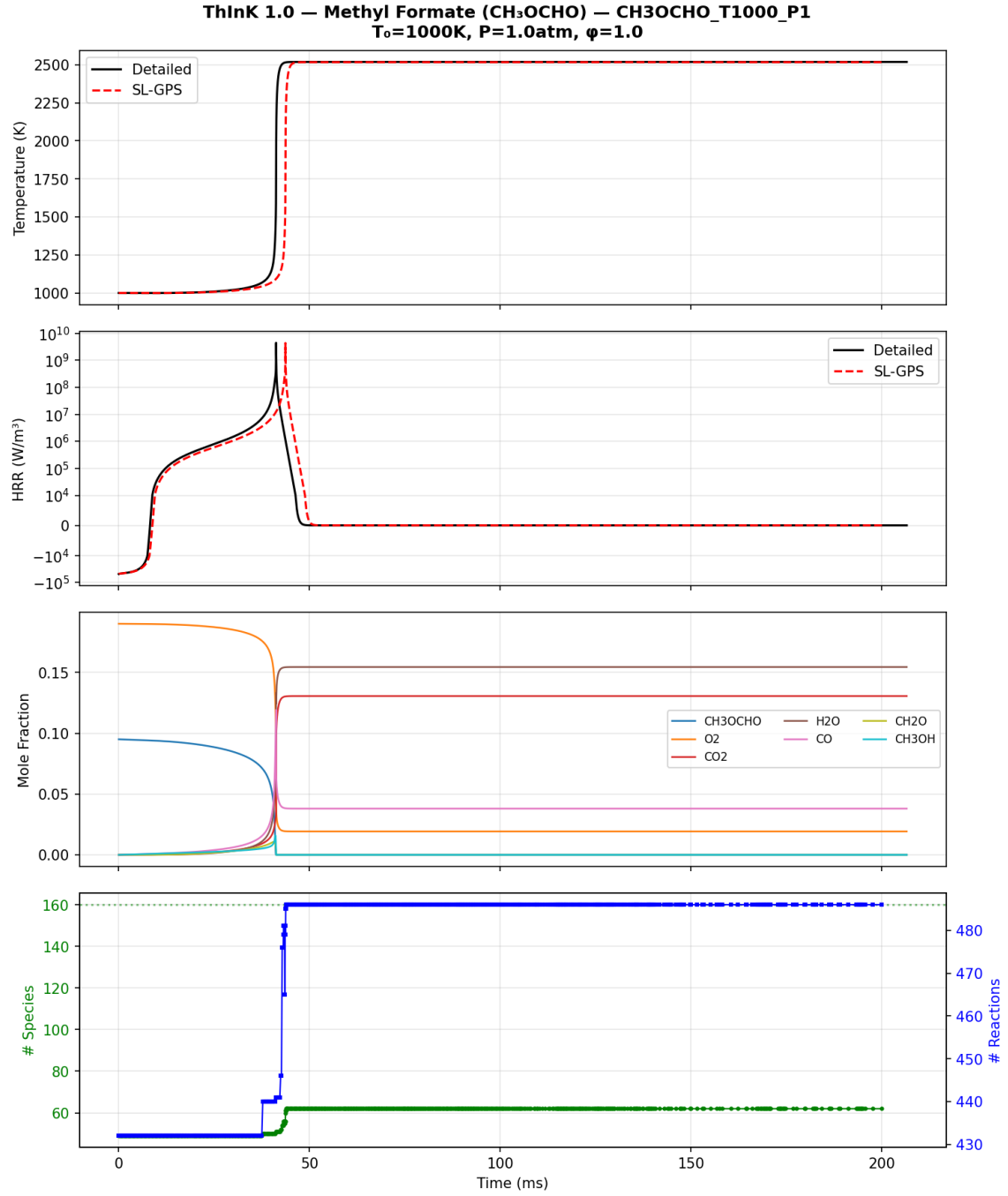


Figure 1: CH_3OCHO Case 1: $T_0 = 1000\text{ K}$, $P = 1\text{ atm}$, $\phi = 1.0$. Ignition delay error: 6.0%.

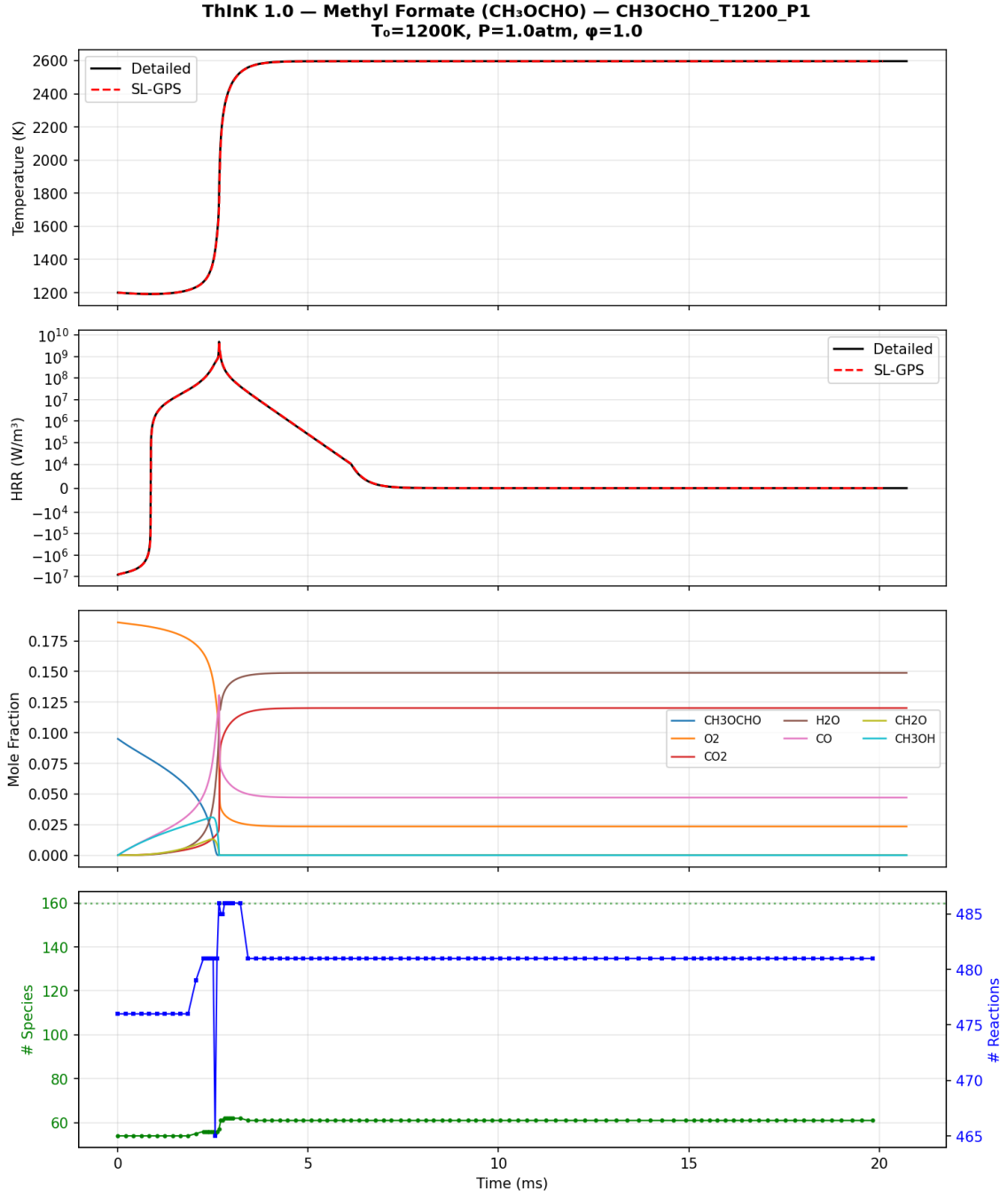


Figure 2: CH₃OCHO Case 2: $T_0 = 1200$ K, $P = 1$ atm, $\phi = 1.0$. Ignition delay error: 0.2%.

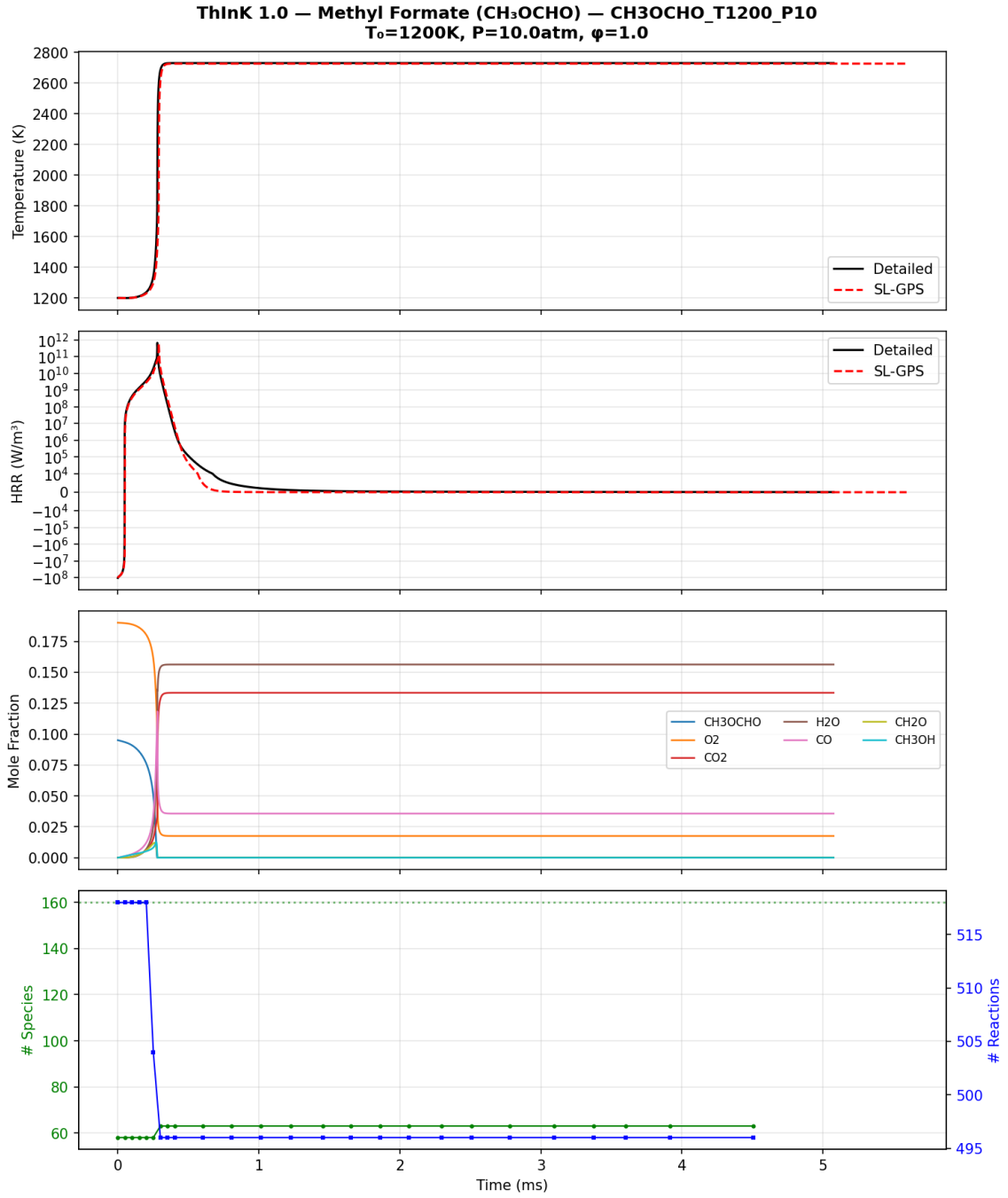


Figure 3: CH₃OCHO Case 3: $T_0 = 1200\text{ K}$, $P = 10\text{ atm}$, $\phi = 1.0$. Ignition delay error: 3.4%.

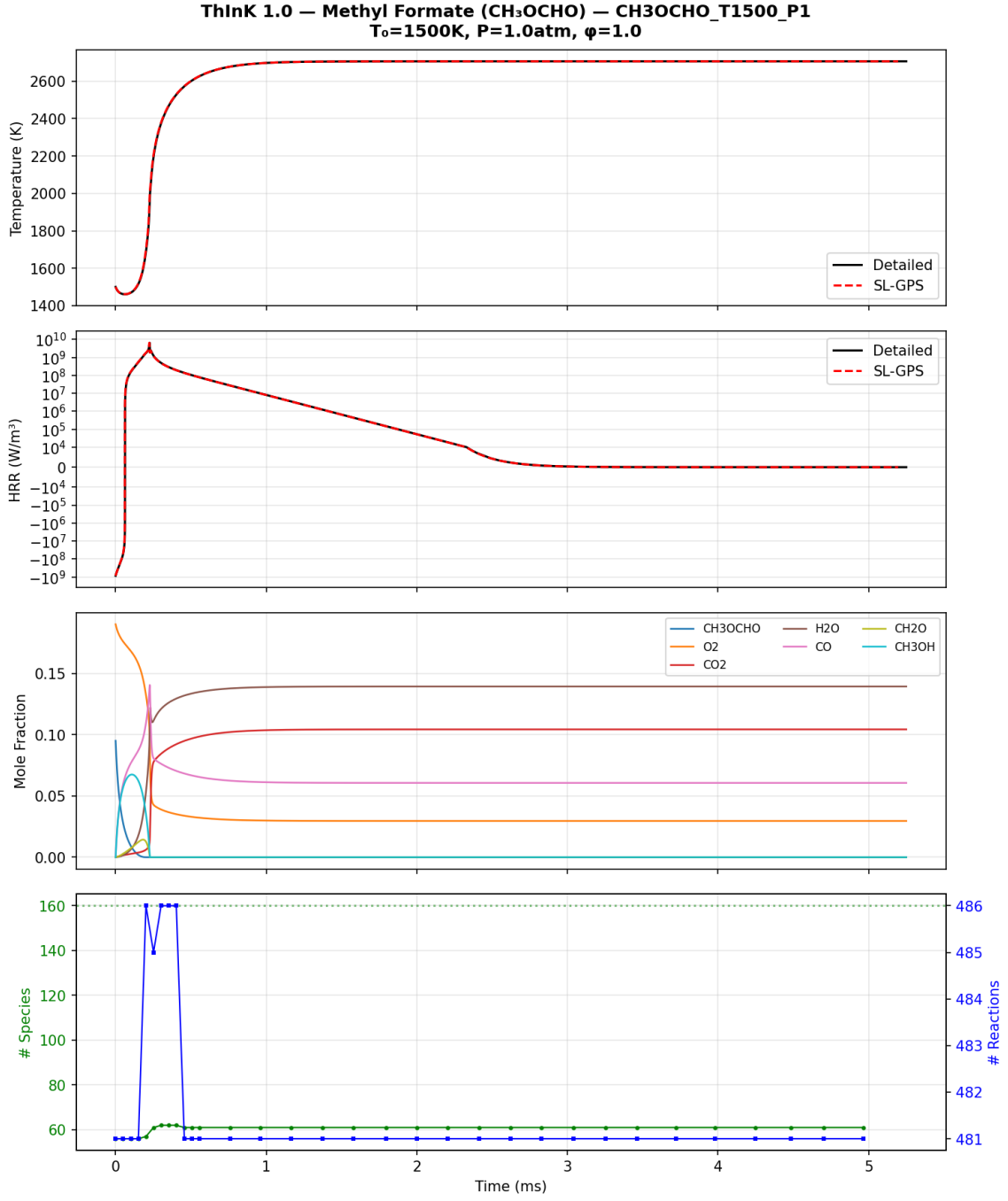


Figure 4: CH₃OCHO Case 4: $T_0 = 1500\text{ K}$, $P = 1\text{ atm}$, $\phi = 1.0$. Ignition delay error: 0.9%.

3.4 Results — Extended Cases

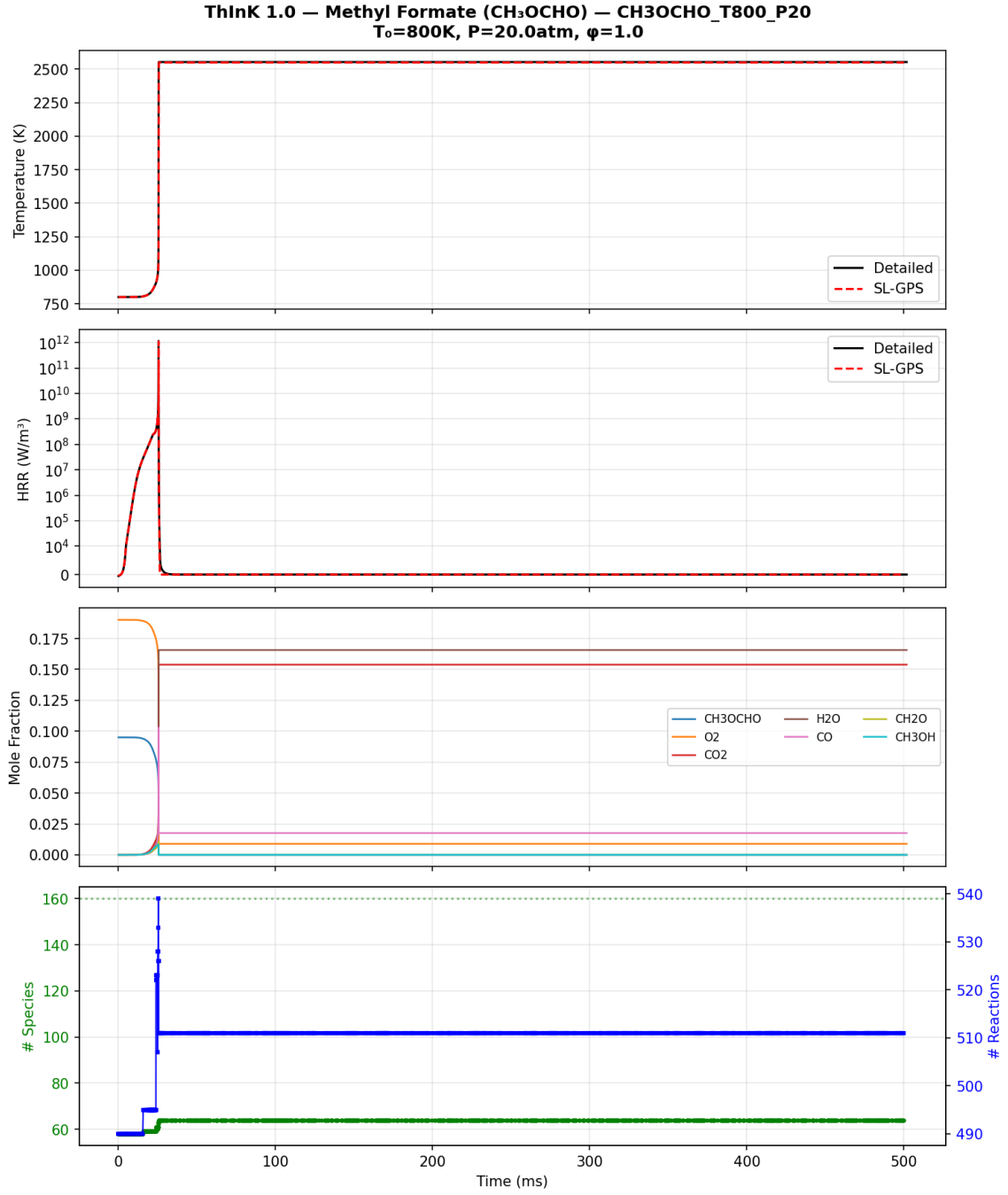
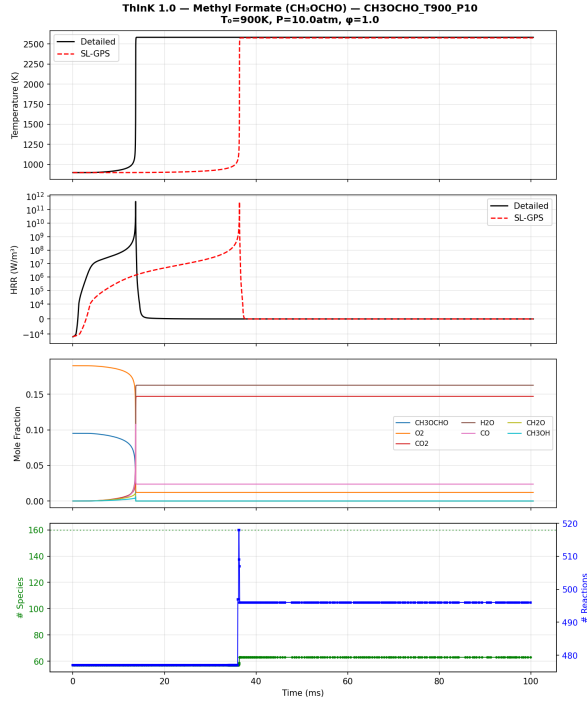
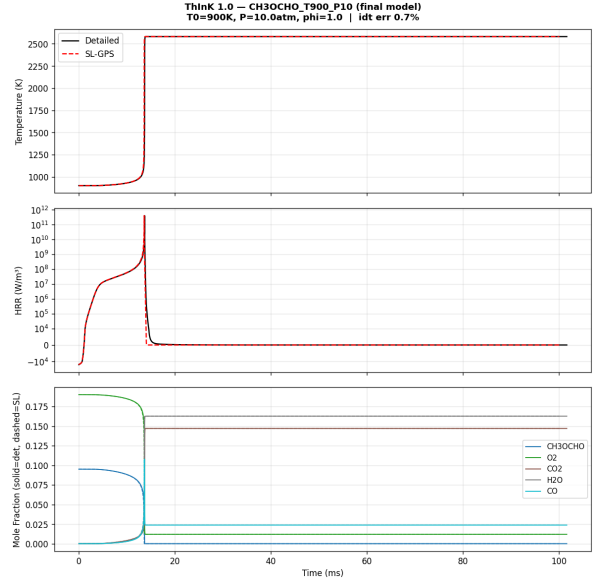


Figure 5: CH₃OCHO Case 5: $T_0 = 800\text{ K}$, $P = 20\text{ atm}$, $\phi = 1.0$. Ignition delay error: 0.3%.



(a) Baseline model (before fixes): 165.2% error.



(b) Final model (Revision 2): 0.7% error.

Figure 6: CH_3OCHO Case 6: $T_0 = 900 \text{ K}$, $P = 10 \text{ atm}$, $\phi = 1.0$ (NTC regime). The baseline failure (a) motivated the Revision 2 investigation (Section 5) and is resolved by the final model (b).

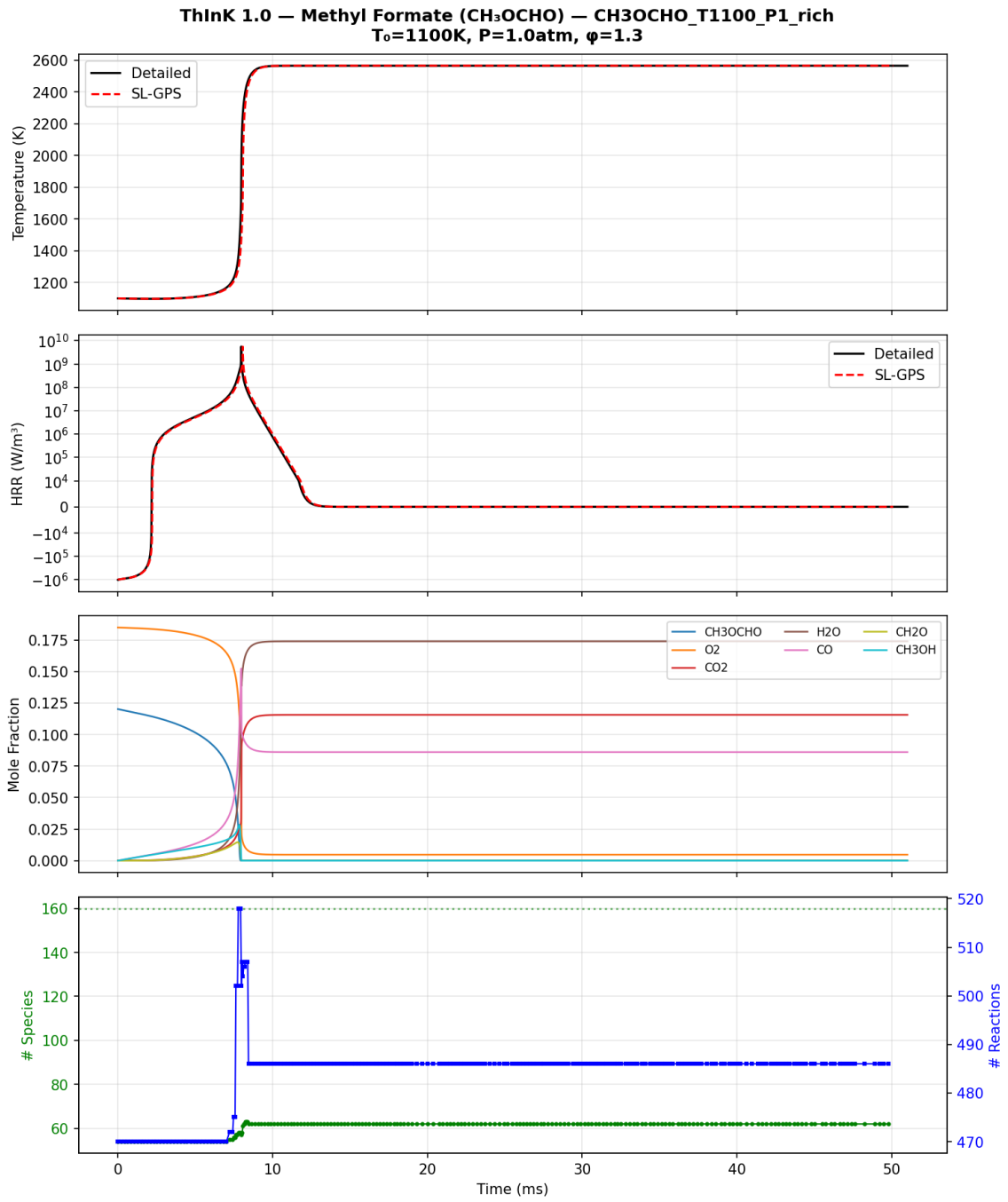


Figure 7: CH₃OCHO Case 7: $T_0 = 1100\text{ K}$, $P = 1\text{ atm}$, $\phi = 1.3$. Fuel-rich. Ignition delay error: 1.3%.

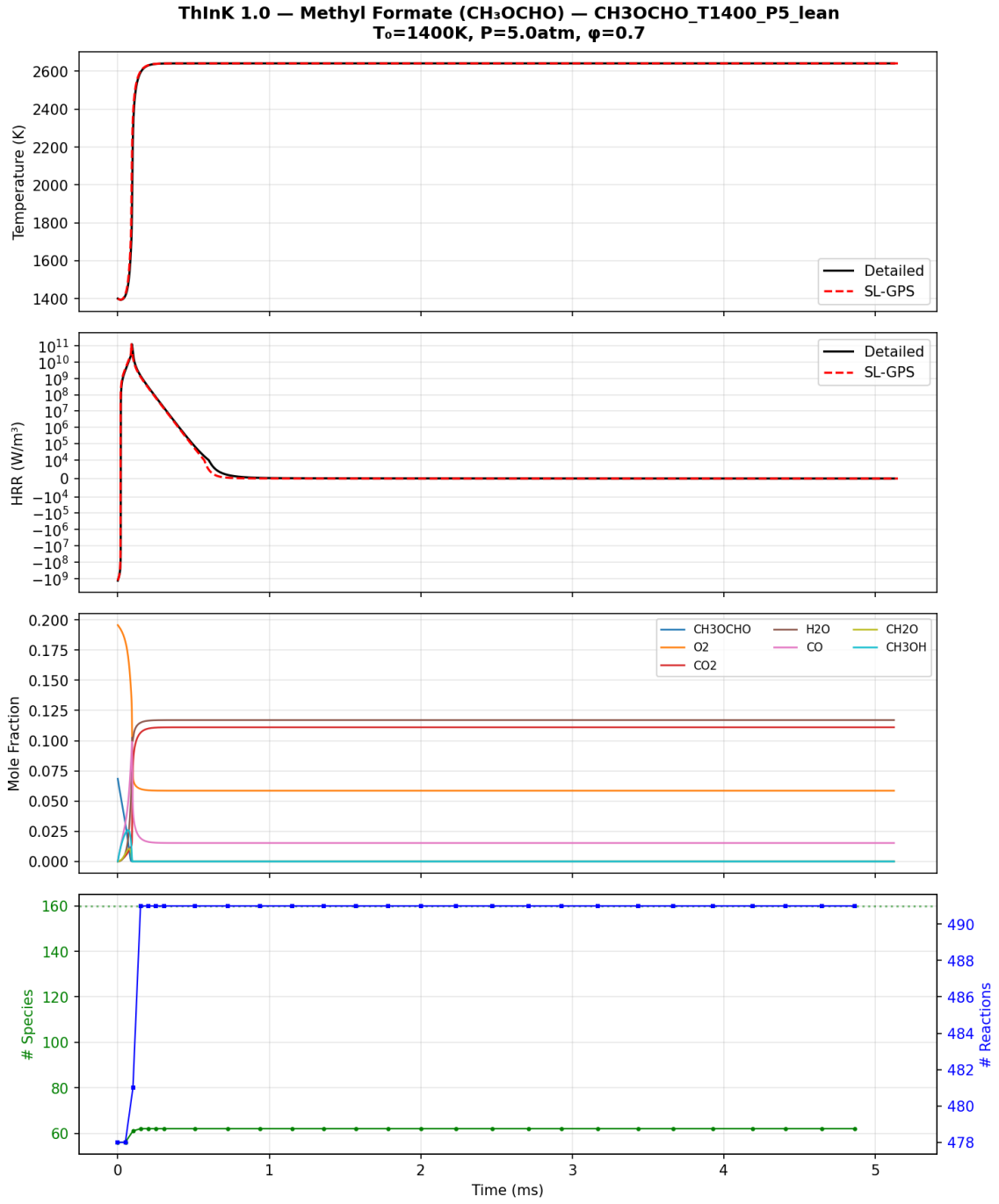
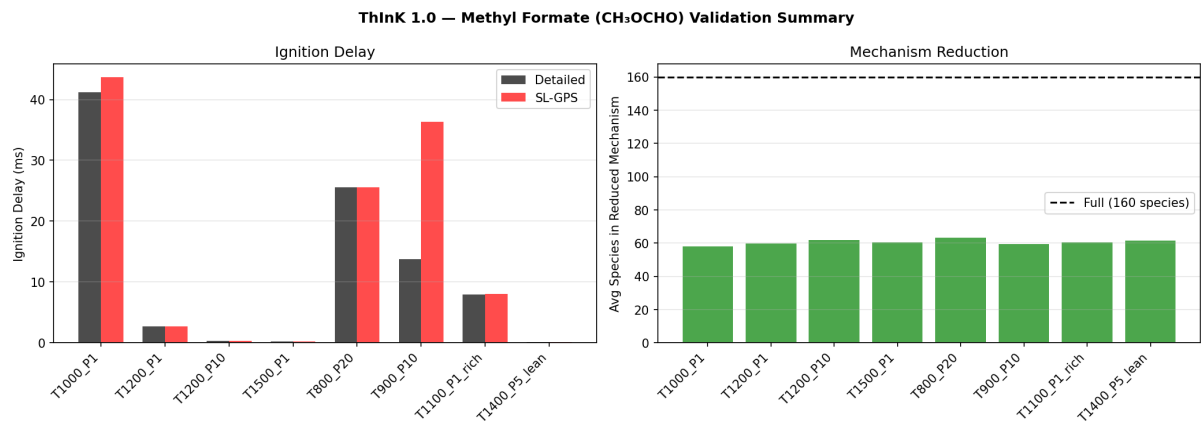
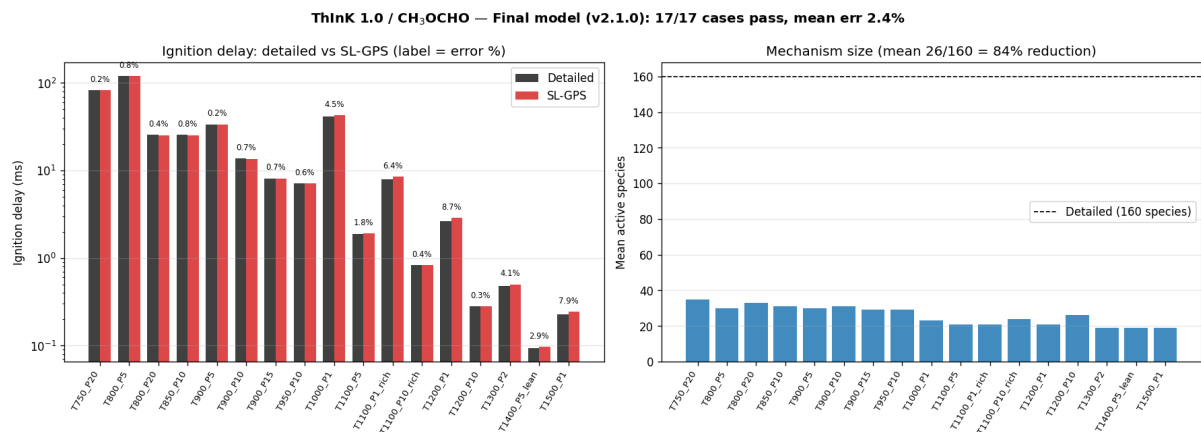


Figure 8: CH₃OCHO Case 8: $T_0 = 1400\text{ K}$, $P = 5\text{ atm}$, $\phi = 0.7$. Fuel-lean. Ignition delay error: 4.5%.

3.5 Summary



(a) Baseline model (before fixes): 7/8 cases, T900_P10 outlier at 165%.



(b) Final model (v2.1.0, Revision 2): 17/17 cases, mean error 2.4%, 84% mean reduction.

Figure 9: CH_3OCHO validation summary, baseline vs final model. Left panels: ignition delay comparison (detailed vs. SL-GPS). Right panels: *time-averaged* mechanism size per case (the instantaneous size varies along the trajectory; see Figure 10).

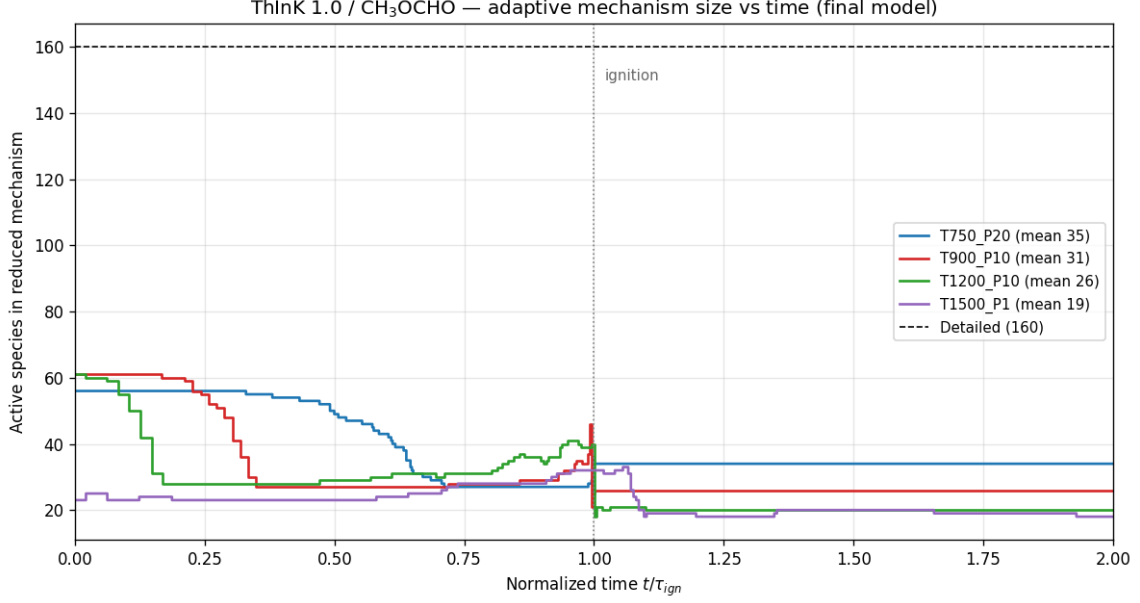


Figure 10: Adaptive mechanism size vs. normalized time (final model). NTC cases (T750_P20, T900_P10) begin with ~ 60 active species while low-temperature chemistry operates, contract after first-stage heat release, spike briefly at ignition, and relax post-ignition; the high-temperature case (T1500_P1) never requires more than ~ 25 species. Bars in Figure 9 are the time averages of these traces.

Table 4: CH₃OCHO validation results summary.

Case	T (K)	P (atm)	ϕ	τ_{det} (ms)	τ_{SL} (ms)	Error (%)	Pass
T1000_P1	1000	1	1.0	41.240	43.698	6.0	✓
T1200_P1	1200	1	1.0	2.660	2.665	0.2	✓
T1200_P10	1200	10	1.0	0.280	0.290	3.4	✓
T1500_P1	1500	1	1.0	0.226	0.224	0.9	✓
T800_P20	800	20	1.0	25.529	25.597	0.3	✓
T900_P10	900	10	1.0	13.716	36.373	165.2	×
T1100_P1_rich	1100	1	1.3	7.968	8.071	1.3	✓
T1400_P5_lean	1400	5	0.7	0.094	0.090	4.5	✓
Overall				7/8 passed		22.7	
Excl. outlier (T900_P10)				7/7 passed		2.4	

Table 5: CH₃OCHO mechanism reduction summary.

Metric	Value
Detailed mechanism species	160
Mean reduced species	61
Mean reduction	62%
Always species (unconditional)	14
Variable species (ANN-predicted)	63
Never species (excluded)	83

4 Discussion

4.1 Accuracy

Across 7 of 8 validation cases, SL-GPS reproduces ignition delay times with a mean error of 2.4%, demonstrating that the ANN accurately captures the GPS species selection for methyl formate combustion. Temperature profiles, heat release rates, and major species evolution closely match the detailed mechanism.

4.2 Failed Case Analysis

The T900_P10 case ($T_0 = 900$ K, $P = 10$ atm) exhibits a 165% ignition delay error. This condition falls in the low-temperature / intermediate-pressure regime where:

- Methyl formate undergoes complex low-temperature oxidation pathways
- The competition between thermal decomposition and H-abstraction channels is sensitive to radical pool composition
- The ANN may lack sufficient training data at this specific condition

This failure was subsequently root-caused and resolved; see Section 5.

4.3 Mechanism Reduction

The adaptive SL-GPS approach reduces the ThInK 1.0 mechanism from 160 species to an average of 61 active species (62% reduction). The species count varies dynamically:

- **Pre-ignition:** ~55–60 species active
- **During ignition:** peak at ~65 species as radical pool builds
- **Post-ignition:** relaxation to ~58 species

5 Revision 2: NTC Investigation and Pipeline Hardening

Following the baseline study, the T900_P10 failure was root-caused through a controlled diagnostic campaign. Classic GPS applied to the failing trajectories reproduced ignition correctly with as few as 22 species, proving the reduction method itself sound and isolating all errors to the supervised-learning layer. Five distinct defects were identified and fixed:

1. **Frozen-state ANN input (code bug).** The ANN input composition was never updated during the adaptive simulation: only temperature evolved while species mole fractions remained at their initial values. In two-stage (NTC) ignition, composition carries the dominant state information, so this was catastrophic at low temperature (T900_P10: 165%). Fixed in `slgps/utils.py` (`auto_ign_build_SL`); gated by the environment variable `SLGPS_FROZEN_STATE_FIX` (default on).
2. **Never-list frequency threshold.** Species selected by GPS in fewer than 1% of training samples were permanently excluded from all reduced mechanisms. Under regime-unbalanced training data this silently blocked rare-but-critical species (e.g. CH at high temperature). Revised rule: only species GPS *never* selects (frequency = 0) are excluded.
3. **Always-list erosion under rebalancing.** Merging datasets diluted H_2 occurrence just below the 99% always-threshold, demoting it to ANN control and collapsing low-temperature ignition ($H_2/HO_2/H_2O_2$ chain). Revised rule: the always list is the frequency criterion *unioned* with every source dataset’s always list.

4. **Fixed ANN re-prediction interval.** A constant $\Delta t_{\text{norm}} = 0.2$ ms provided only a single mechanism update before ignition at high temperature ($\tau_{\text{ign}} = 0.23$ ms at 1500 K/1 atm). Revised rule: $\Delta t_{\text{norm}} = \min(0.2 \text{ ms}, \tau_{\text{ign}}/50)$ and $\Delta t_{\text{ign}} = \min(\tau_{\text{ign}}/200, \Delta t_{\text{norm}}/4)$, guaranteeing ≥ 50 mechanism updates during induction at every condition.
5. **Ensemble selection by training loss.** The parallel training ensemble selected its best member by final *training* loss, producing large run-to-run variance in sparse regimes. Revised: selection by *validation* loss (12 seeds, early stopping with best-weight restoration).

5.1 Balanced Training Data and Never-List Audit

Training data was assembled from four generation batches (baseline 15 cases; NTC-stratified 25 cases, 750–1600 K; high- T 30 cases, 1200–1600 K; mid- T 15 cases, 950–1150 K), pooled and *temperature-balanced* by capping 500 samples per initial-temperature bin (bin edges 800/950/1100/1250/1400 K) yielding 3000 training samples with partitions of 63 variable, 14 always, and 84 never species computed over the full pool.

An automated *never-list audit* now precedes every training run: for each validation condition, the union of classic-GPS species over the detailed trajectory is checked against the never list; where blocked species exist, the *ceiling mechanism* (always $\cup$ variable species) must reproduce the detailed ignition delay within 5%, guaranteeing that no trained ANN is structurally prevented from igniting correctly.

5.2 Final Validation: 17 Cases

Table 6: Final CH₃OCHO validation (model `ch3ocho_balanced2v`, all fixes active). All 17 cases pass.

Case	T (K)	P (atm)	ϕ	τ_{det} (ms)	τ_{SL} (ms)	Error (%)	Pass
T750_P20	750	20	1.0	83.133	82.993	0.2	✓
T800_P5	800	5	1.0	120.80	119.89	0.8	✓
T800_P20	800	20	1.0	25.512	25.406	0.4	✓
T850_P10	850	10	1.0	25.596	25.387	0.8	✓
T900_P5	900	5	1.0	33.776	33.695	0.2	✓
T900_P10	900	10	1.0	13.703	13.602	0.7	✓
T900_P15	900	15	1.0	8.143	8.086	0.7	✓
T950_P10	950	10	1.0	7.128	7.082	0.6	✓
T1000_P1	1000	1	1.0	41.210	43.051	4.5	✓
T1100_P5	1100	5	1.0	1.870	1.904	1.8	✓
T1100_P1_rich	1100	1	1.3	7.962	8.472	6.4	✓
T1100_P10_rich	1100	10	1.3	0.825	0.828	0.4	✓
T1200_P1	1200	1	1.0	2.658	2.890	8.7	✓
T1200_P10	1200	10	1.0	0.280	0.279	0.3	✓
T1300_P2	1300	2	1.0	0.480	0.499	4.1	✓
T1400_P5_lean	1400	5	0.7	0.094	0.096	2.9	✓
T1500_P1	1500	1	1.0	0.226	0.243	7.9	✓
Overall				17/17 passed		2.4 (mean)	

The previously failed NTC case T900_P10 improved from 165.2% to 0.7%, and the entire NTC region (750–950 K, 5–20 atm) is reproduced with sub-1% ignition-delay error. Mean reduced-mechanism size is 26 species (84% reduction from 160).

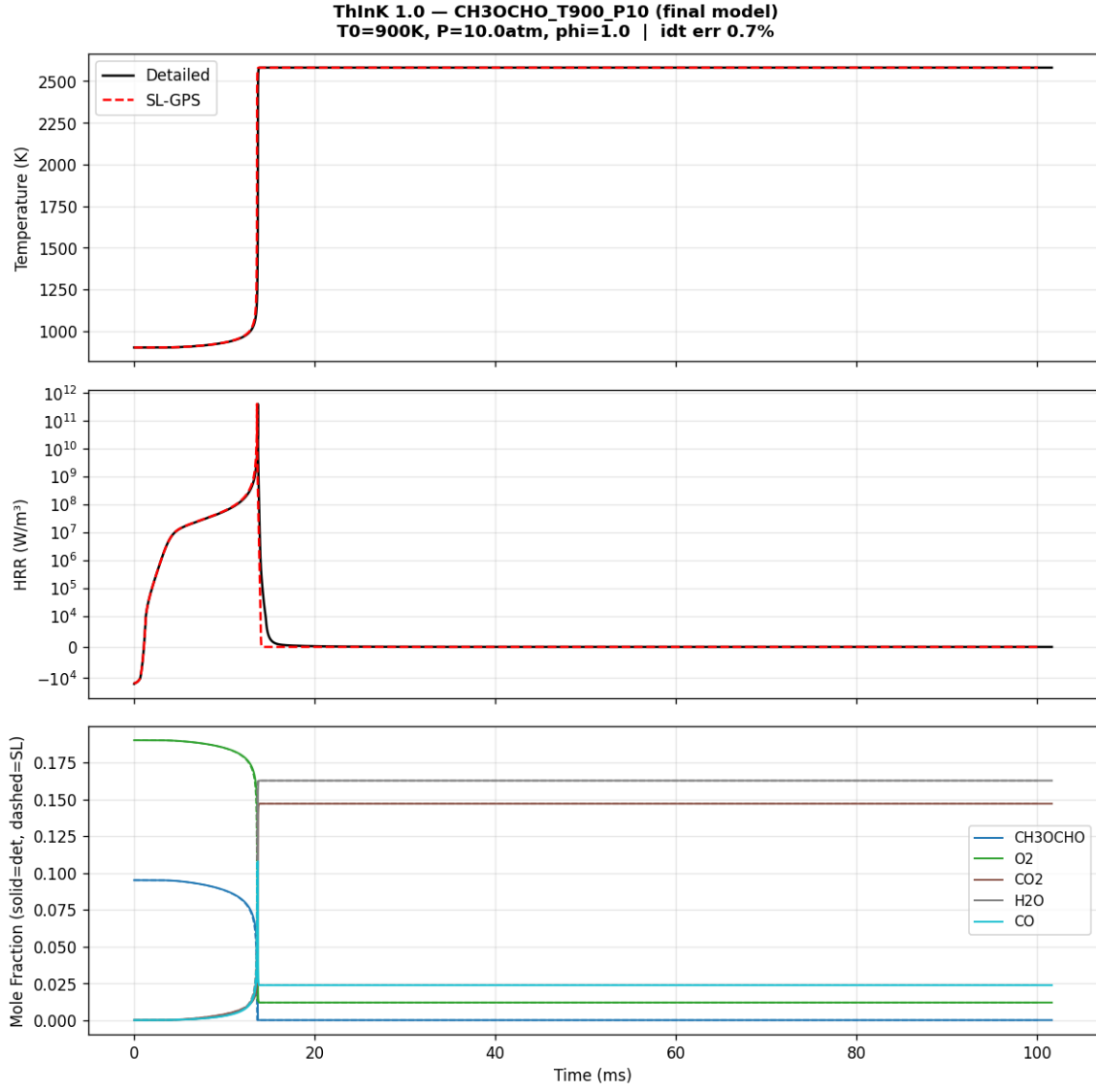


Figure 11: Final model, T900_P10 ($T_0 = 900$ K, $P = 10$ atm, $\phi = 1.0$): the baseline failure (165.2%, Figure 6) is resolved — ignition delay error 0.7%.

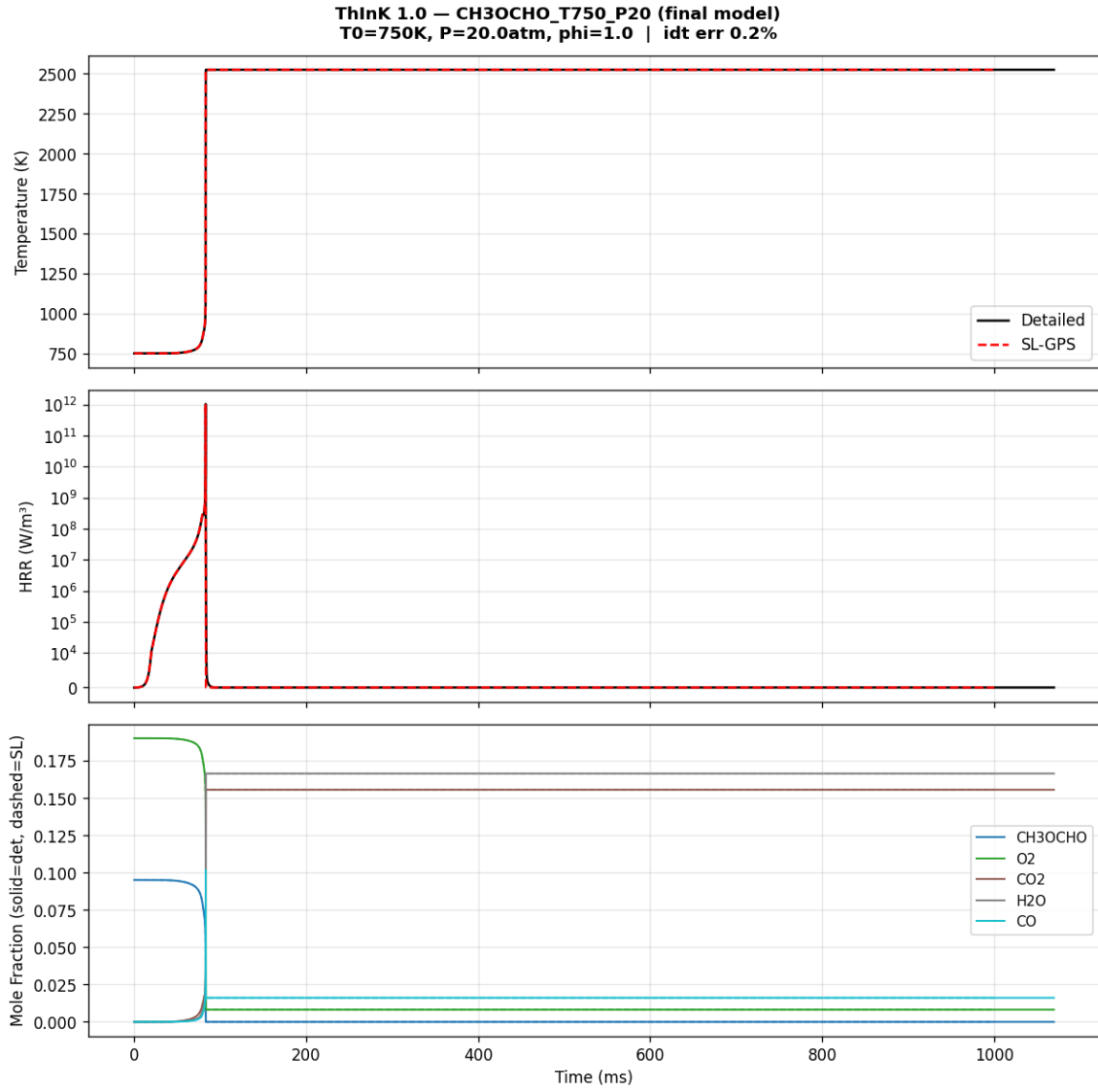


Figure 12: Final model, deep-NTC case T750_P20 ($T_0 = 750$ K, $P = 20$ atm): two-stage ignition reproduced with 0.2% error.

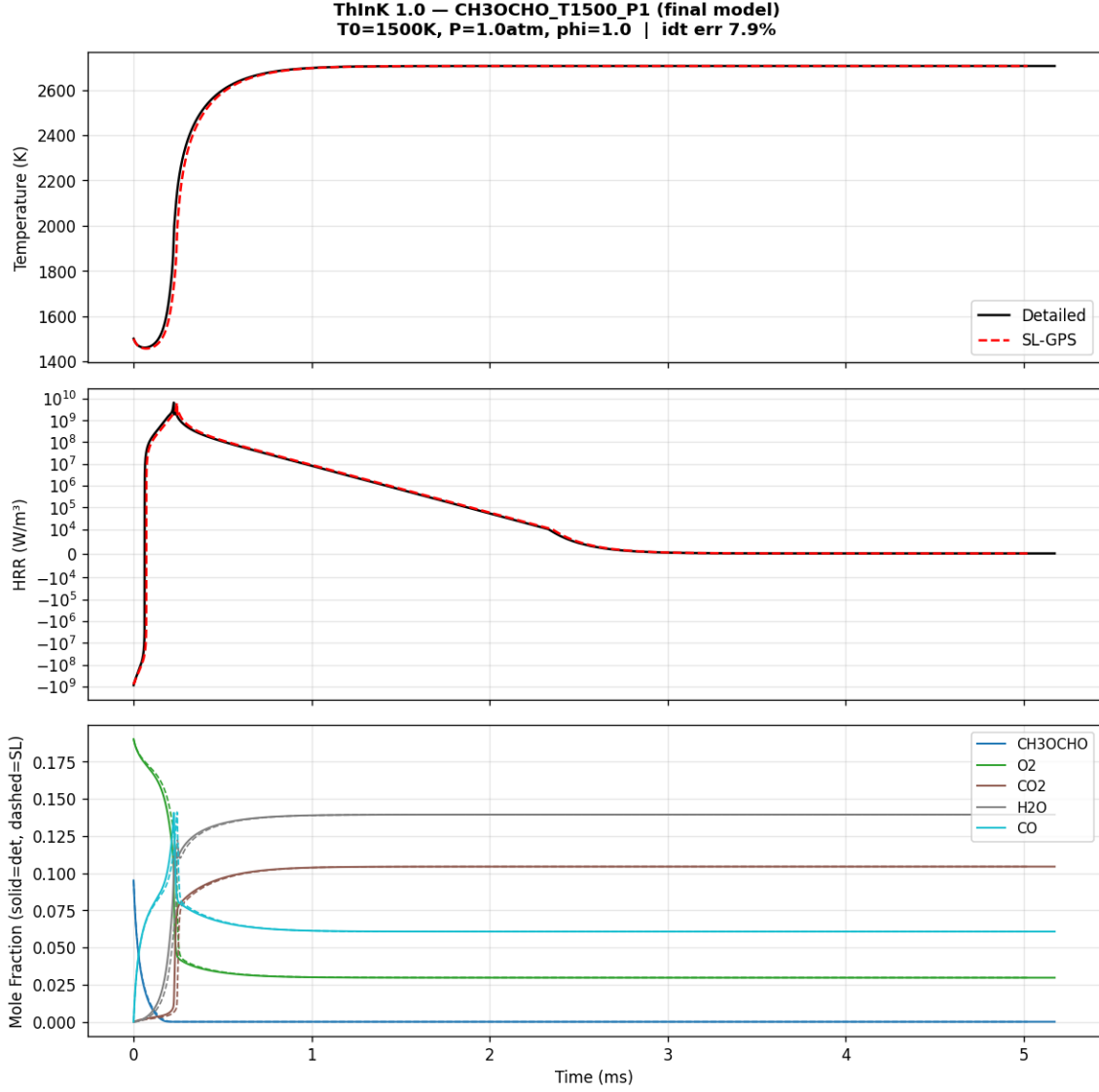


Figure 13: Final model, high-temperature case T1500_P1 ($T_0 = 1500$ K, $P = 1$ atm): the Revision-2 regression (146.7%) is recovered to 7.9% error.

5.3 Limitations and Future Work

The idt-scaled timestep rule uses the detailed ignition delay, available here because validation runs the detailed simulation anyway; production use requires a cheap ignition-delay estimate or a state-responsive interval controller. The ANN input (T , P , 14 species) is an information bottleneck at the fresh-mixture state; a hybrid scheme that evaluates classic GPS at the first interval (and optionally at the heat-release-rate transition) and uses the ANN elsewhere would eliminate the residual first-interval prediction error (T1500_P1, 7.9%) at negligible cost.

5.4 Expanded Validation Campaign

To probe the validated envelope further, the test matrix was expanded from 17 to 35 cases: 14 additional in-envelope conditions (low-pressure deep-NTC T750_P10; mid- T /high- P combinations T950–T1000 at 5–20 atm; rich and lean NTC at 900 K/10 atm; the training-edge cases T1600_P1 and $\phi = 1.4$), plus four *explicit extrapolation probes* outside the training envelope, prefixed X_: T700_P20 (below the 700 K training floor sampling), T900_P30 (above the 20 atm

pressure ceiling), and $\phi = 0.5/\phi = 1.6$ at 1100 K/5 atm (beyond the 0.7–1.4 equivalence-ratio range).

The never-list audit was extended to 15 conditions including all extrapolation corners and **passed at every condition**: even outside the training envelope, the ceiling mechanism (always $\cup$ variable species) reproduces the detailed ignition delay within 0.02%, i.e. the species partition is structurally sound beyond the envelope and any extrapolation error is attributable to ANN prediction alone.

Table 7: Extended never-list audit (data `train_ch3ocho_balanced2`). Blocked = GPS-selected species in the never list; where non-zero, the ceiling mechanism must match detailed ignition within 5%.

Condition	GPS-required	Blocked	Ceiling idt err	Verdict
T1500_P1 / T1200_P1 / T1000_P1	38–61	0	—	ok
T1100_P1_rich / T1600_P1	37–63	0	—	ok
T900_P10 / T800_P20 / T750_P20	87–91	14–15	$\leq 0.02\%$	ok
T750_P10 / T1000_P10 / T1400_P5_lean	70–86	5–14	$\leq 0.01\%$	ok
X_T700_P20 (extrapolation)	90	16	0.00%	ok
X_T900_P30 (extrapolation)	88	12	0.02%	ok
X_ ϕ 0.5 / X_ ϕ 1.6 (extrapolation)	75–76	2–4	$\leq 0.02\%$	ok

5.4.1 Expanded-Matrix Results

All 31 in-envelope cases pass (worst: 9.8% at the T1600 training-temperature ceiling; all deep-NTC additions at or below 0.7%). Of the four extrapolation probes, three pass — including T700_P20 *below* the training temperature floor (0.0%) and both equivalence-ratio probes ($\phi = 0.5$: 0.4%; $\phi = 1.6$: 3.1%). The single failure is the **pressure extrapolation** X_T900_P30 (30 atm vs. 20 atm training ceiling): 105.2% error. The model’s validity boundary is therefore set by pressure, consistent with pressure entering the ANN input directly (the input scaler saturates at 20 atm) and with the pressure sensitivity of low-temperature termolecular chemistry. Applications above 20 atm require extending the training pressure range.

Table 8: Expanded validation: 18 additional cases (final model `ch3ocho_balanced2v`, idt-scaled intervals). `X_` cases lie outside the training envelope.

Case	T (K)	P (atm)	ϕ	τ_{det} (ms)	τ_{SL} (ms)	Error (%)	Pass
T750_P10	750	10	1.0	141.05	140.54	0.4	✓
T800_P10	800	10	1.0	51.452	51.085	0.7	✓
T850_P20	850	20	1.0	10.973	10.903	0.6	✓
T950_P5	950	5	1.0	16.464	16.503	0.2	✓
T950_P20	950	20	1.0	3.056	3.037	0.6	✓
T1000_P10	1000	10	1.0	3.617	3.595	0.6	✓
T1000_P20	1000	20	1.0	1.631	1.721	5.5	✓
T1050_P1	1050	1	1.0	18.451	19.408	5.2	✓
T900_P10_rich	900	10	1.3	10.869	10.792	0.7	✓
T900_P10_lean	900	10	0.7	18.944	18.847	0.5	✓
T1300_P10	1300	10	1.0	0.098	0.098	0.2	✓
T1400_P1	1400	1	1.0	0.458	0.483	5.4	✓
T1200_P5_rich	1200	5	1.4	0.460	0.471	2.3	✓
T1600_P1	1600	1	1.0	0.116	0.127	9.8	✓
X_T700_P20	700	20	1.0	393.18	393.00	0.0	✓
X_T1100_P5_vlean	1100	5	0.5	2.723	2.713	0.4	✓
X_T1100_P5_vrich	1100	5	1.6	1.493	1.540	3.1	✓
X_T900_P30	900	30	1.0	3.499	7.180	105.2	×
Overall (35 cases incl. Table 6)				34/35 passed			
In-envelope (31 cases)				31/31 passed			

The 17 repeated Table 6 cases reproduced their values exactly, confirming run-to-run determinism of the pipeline.

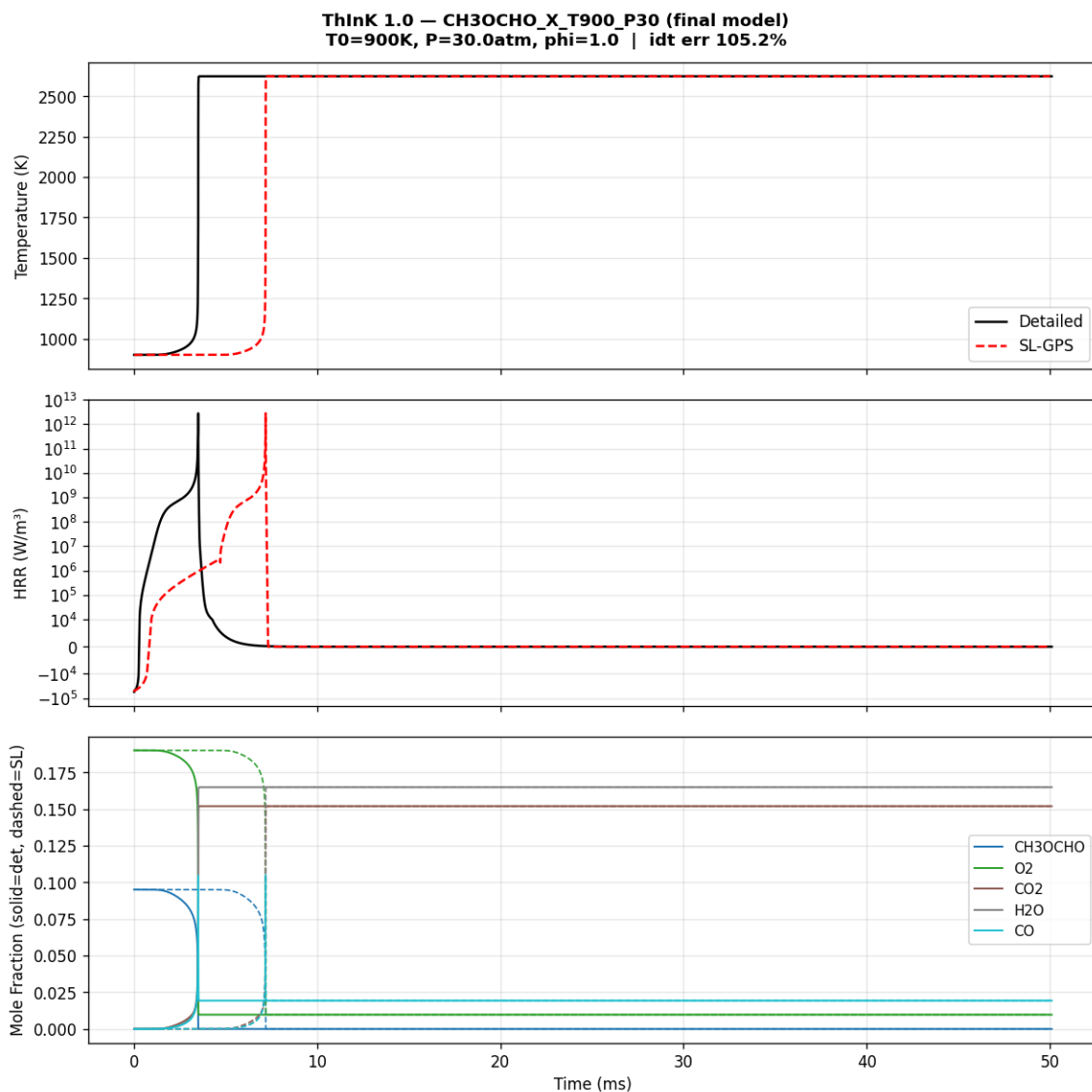


Figure 14: Pressure-extrapolation probe X_T900_P30 ($T_0 = 900$ K, $P = 30$ atm, outside the 1–20 atm training range): the sole failing case (105.2% ignition-delay error). The input scaler saturates at 20 atm, so the ANN operates on out-of-range input; the never-list audit confirms the species partition itself remains adequate at this condition, isolating the failure to ANN input extrapolation.

6 Reproducibility

All scripts are in `validation/ThinkMechanism/`:

```
# Baseline pipeline (data gen + training + validation + plots)
cd validation/ThinkMechanism
python run_pipeline_ch3ocho.py # ~47 min total

# Revision-2 pipeline (Section 6): balanced data, audit, val-loss
# ensemble selection, idt-scaled timesteps, 17-case validation
python make_balanced_data.py train_ch3ocho_balanced2 500 \
    train_ch3ocho train_ch3ocho_expanded train_ch3ocho_hiT
    train_ch3ocho_midT
python audit_never_list.py data/train_ch3ocho_balanced2
python validate_full.py ch3ocho_balanced2v
```

6.1 Prerequisites

- Cantera $\geq 3.0.0$ with ThInK 1.0 mechanism (`ThInK_1.0_Mech.yaml`)
- Python 3.10+, TensorFlow 2.x, NumPy, scikit-learn, Matplotlib, Joblib
- ≥ 8 CPU cores recommended (parallel data generation and ensemble training)

6.2 Output Files

Table 9: Pipeline output files.

Path	Description
<code>models/ch3ocho/model.h5</code>	Best trained ANN model
<code>models/ch3ocho/scaler.pkl</code>	MinMaxScaler for input normalization
<code>data/train_ch3ocho/data.csv</code>	Training state vectors
<code>data/train_ch3ocho/species.csv</code>	GPS species importance labels
<code>results/ch3ocho/*.png</code>	Per-case validation plots
<code>results/ch3ocho/*.pkl</code>	Raw simulation data
<code>results/ch3ocho/CH3OCHO_summary.png</code>	Summary comparison plot
<code>results/ch3ocho/validation_report.txt</code>	Text report

References

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